

Revisiting a Published NLP Study: Do Transformers Beat Classical ML on Small Datasets?

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Abstract

This report revisits a previously published study (Kobra et al., 2024) that classified the impact of COVID-19 on Bangladeshi students’ academic, mental, and social well-being using eight classical machine learning algorithms and five deep learning architectures, achieving accuracies of up to 98.75%. While the original study benchmarked its results against transformer-based studies from the literature, those comparisons used different datasets, leaving an open question: would a transformer applied directly to *this* dataset actually outperform the original models? We fine-tune BanglaBERT, a pretrained Bengali-language transformer, separately on each of the three classification tasks, and evaluate it using 5-fold stratified cross-validation for a statistically robust comparison. Across all three tasks, BanglaBERT does not outperform the original study’s best classical ML models, and performs roughly on par with (slightly below) the original deep learning architectures. We argue this is a genuinely informative negative result: with only 320 training examples per fold, a large pretrained transformer cannot leverage its capacity advantage over simpler models well-suited to small, clean datasets — a finding consistent with overfitting behavior already observed in the original study’s own deep learning models.

1 Introduction

Kobra et al. (2024) [1] published a study examining the psychological and academic impact of the COVID-19 pandemic on 400 Bangladeshi students, using self-administered survey responses labeled as Positive or Negative sentiment across three life-impact dimensions: academic life, mental health, and social life. The study compared eight classical machine learning algorithms (including SGD, KNN, Random Forest, and others) against five deep learning architectures (LSTM, BiLSTM, CNN, and two hybrid models), achieving accuracies ranging from 92.50% to 98.75% depending on the model and dimension.

Table 12 of the original paper situates these results against transformer-based approaches reported in other published studies — but those benchmarks were computed on *different* datasets entirely, making the comparison indicative at best. No transformer had, to our knowledge, actually been fine-tuned and evaluated on this specific 400-response dataset.

This report closes that gap. We fine-tune BanglaBERT [2], a BERT-family model pretrained on Bengali text, directly on the same dataset and the same three classification tasks used in the original study, and compare the results under a like-for-like evaluation protocol.

2 Dataset

The dataset consists of 400 survey responses collected from Bangladeshi students aged 10 and above, via self-administered questionnaires. Each response contains three free-text fields — `impact_on_academic_life`, `impact_on_mental_health`, and `impact_on_social_life` — and a single shared sentiment label (`Positive` or `Negative`) applied consistently across all three fields for that respondent. The dataset is perfectly balanced (200 Positive, 200 Negative), with no missing values or duplicate rows. Text is written in Bengali.

Critically, the same label is the classification target for all three text columns: the original study effectively trained three separate models per architecture, one per text column, each predicting the same binary outcome from a different piece of text.

3 Methodology

3.1 Model

We use `sagorsarker/bangla-bert-base`, a BERT-base model pretrained on Bengali text, available via the Hugging Face `transformers` library. This choice was made for practical reliability: it requires no custom text normalization pipeline and integrates directly with the standard `AutoTokenizer` / `AutoModelForSequenceClassification` interface.

3.2 Fine-Tuning Setup

For each of the three text columns, BanglaBERT was fine-tuned independently as a binary sequence classifier:

- Tokenization: max sequence length 64 tokens, padded/truncated
- Epochs: 4
- Batch size: 16 (train and eval)
- Learning rate: 2×10^{-5}
- Optimizer: AdamW (Hugging Face `Trainer` default)

3.3 Evaluation Protocol

An initial evaluation used a single 80/20 train/test split. However, with only 80 examples in the test set, a single split is statistically fragile — one misclassified example shifts accuracy by 1.25 percentage points. We therefore adopt **5-fold stratified cross-validation** as the primary evaluation protocol: for each fold, a fresh BanglaBERT model is fine-tuned from the pretrained checkpoint (to prevent information leakage between folds) on 4/5 of the data and evaluated on the held-out fold. We report the mean and standard deviation of accuracy and F1 score across the 5 folds.

We note that the original study’s own evaluation used a single train/test split without cross-validation; our protocol is therefore methodologically more rigorous than the baseline we compare against.

4 Results

Table 1 reproduces the best-performing classical ML and deep learning results from the original study (Tables 5 and 6 of Kobra et al., 2024), which serve as our comparison baselines.

Table 1: Original study baselines (Kobra et al., 2024)

Column	Best Classical ML	Best Deep Learning
Academic	SGD — 95.00%	BiLSTM — 92.50%
Mental Health	KNN — 93.75%	LSTM — 98.75%
Social	SGD / Multi-NB — 95.00%	BiLSTM / CNN / CNN-LSTM — 92.50%

Table 2 shows our 5-fold cross-validated BanglaBERT results alongside these baselines.

Table 2: BanglaBERT (5-fold CV) vs. original study baselines

Column	BanglaBERT (mean $\pm$ std, %)	Best ML (%)	Best DL (%)
Academic	91.75 $\pm$ 1.70	95.00	92.50
Mental Health	90.75 $\pm$ 1.27	93.75	98.75
Social	90.75 $\pm$ 3.32	95.00	92.50

Per-fold accuracy for transparency:

- **Academic:** 88.8%, 93.8%, 92.5%, 91.2%, 92.5%
- **Mental Health:** 92.5%, 91.2%, 91.2%, 90.0%, 88.8%
- **Social:** 92.5%, 92.5%, 95.0%, 86.2%, 87.5%

BanglaBERT did not outperform the original study’s best classical ML model on any of the three tasks. It performed roughly on par with, or slightly below, the original deep learning architectures on two of three tasks (Academic, Social), and clearly below on Mental Health, where the original LSTM’s 98.75% result leaves very little room for any model to improve upon.

5 Discussion

The consistent 3–5 percentage point gap between BanglaBERT and the classical ML baselines — larger than the observed standard deviations in two of three tasks — suggests a real effect rather than statistical noise. We interpret this as a demonstration of a well-established but frequently overlooked principle in applied NLP: *a large pretrained transformer requires a meaningful volume of task-specific fine-tuning data to leverage its capacity advantage*. With only 320 training examples per fold, BanglaBERT (110M+ parameters) could not out-learn classical models such as SGD and KNN, which are comparatively well-suited to small, clean, well-labeled datasets.

This finding is not isolated. The original study’s own Table 9 reports *negative* R^2 values for its CNN and CNN-LSTM models on this same dataset — an independent signal that model complexity was already outrunning the available data before a transformer was introduced. Our result extends this observation to a substantially larger pretrained model.

The Social column shows the highest cross-fold variance (± 3.32), with fold accuracies ranging from 86.2% to 95.0% — an 8.8-point spread. We interpret this as evidence that BanglaBERT’s

performance on this particular task is comparatively less stable, plausibly due to greater semantic overlap between positive and negative responses in the social-life text column, though we do not have direct evidence to confirm this hypothesis and flag it as a direction for further investigation rather than a settled conclusion.

6 Limitations

- **Dataset size.** At 400 total examples, this dataset is small even by the standards of typical fine-tuning studies; our conclusions are specific to this scale and should not be generalized to larger Bengali NLP tasks.
- **Single model architecture.** We evaluate only `sagorsarker/bangla-bert-base`; other Bengali transformers (e.g., BanglaBERT from CSEBUETNLP, XLM-RoBERTa) were not tested and may behave differently.
- **Fixed hyperparameters.** We did not perform hyperparameter search (learning rate, epochs, sequence length); a more extensively tuned transformer configuration might narrow or close the observed gap.
- **Practical subsampling.** Each fold effectively fine-tunes on 256 training examples (4/5 of 320); this is near the lower bound of what fine-tuning literature typically considers viable for transformer adaptation.

7 Conclusion

Fine-tuning BanglaBERT directly on the same dataset used in a previously published study did not yield an improvement over the original classical ML or deep learning results. Cross-validated evaluation confirms this is a stable finding rather than an artifact of a single unlucky data split. Rather than a negative outcome, we view this as a practically useful result: for small, well-curated survey-style NLP datasets of this scale, classical machine learning remains a legitimate — and often preferable — first choice over reaching for a transformer by default. Future work could test this boundary directly by artificially varying training set size to identify the approximate data volume at which a transformer’s advantage begins to emerge on this task.

References

- [1] Kobra, K., Sarkar Sammi, S., Rahman, N., Khushbu, S. A., & Islam, M. (2024). Multihead Text Mining from COVID-19 Feedback Using Machine Learning, Deep Learning, and Hybrid Deep Learning Approaches. *Journal of Sensors*, 2024, Article ID 3027199. <https://doi.org/10.1155/2024/3027199>
- [2] Sarker, S. (2020). BanglaBERT: Bengali Pretrained Language Model. <https://huggingface.co/sagorsarker/bangla-bert-base>